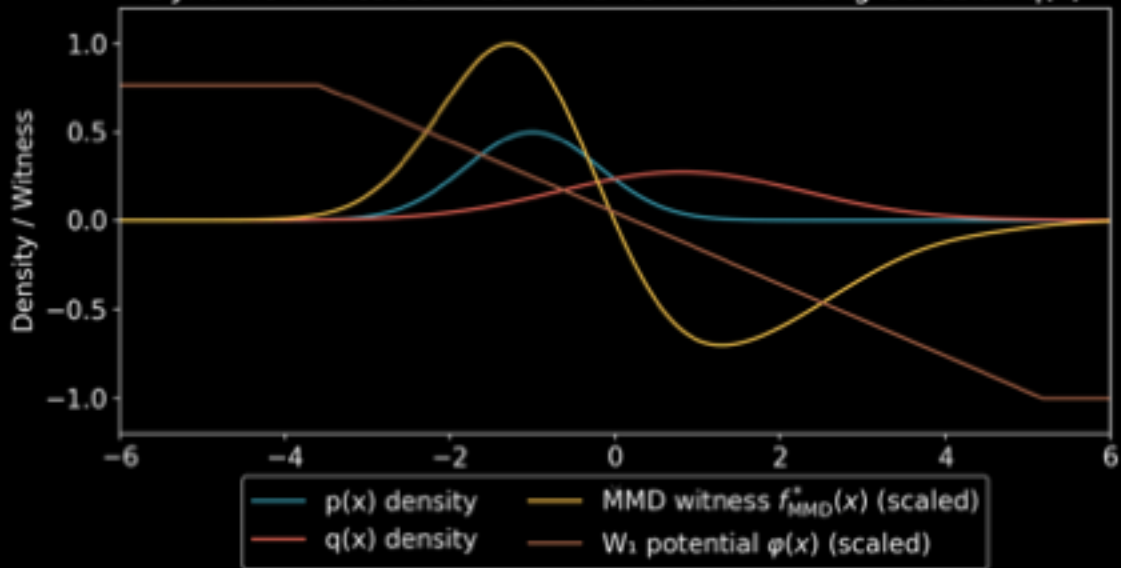
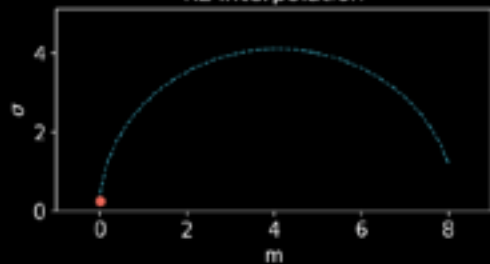


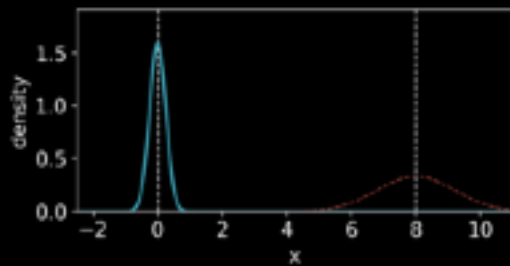
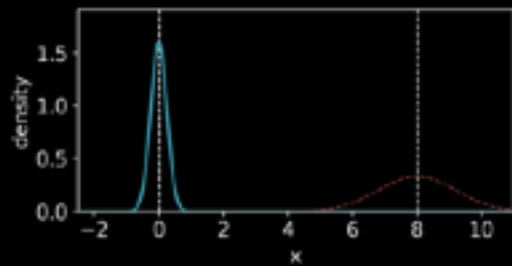
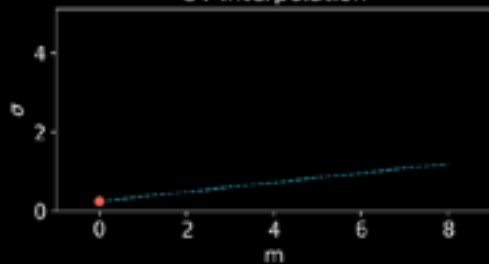
Dynamic Witnesses in 1D: MMD vs W_1 for drifting Gaussian $q(x)$



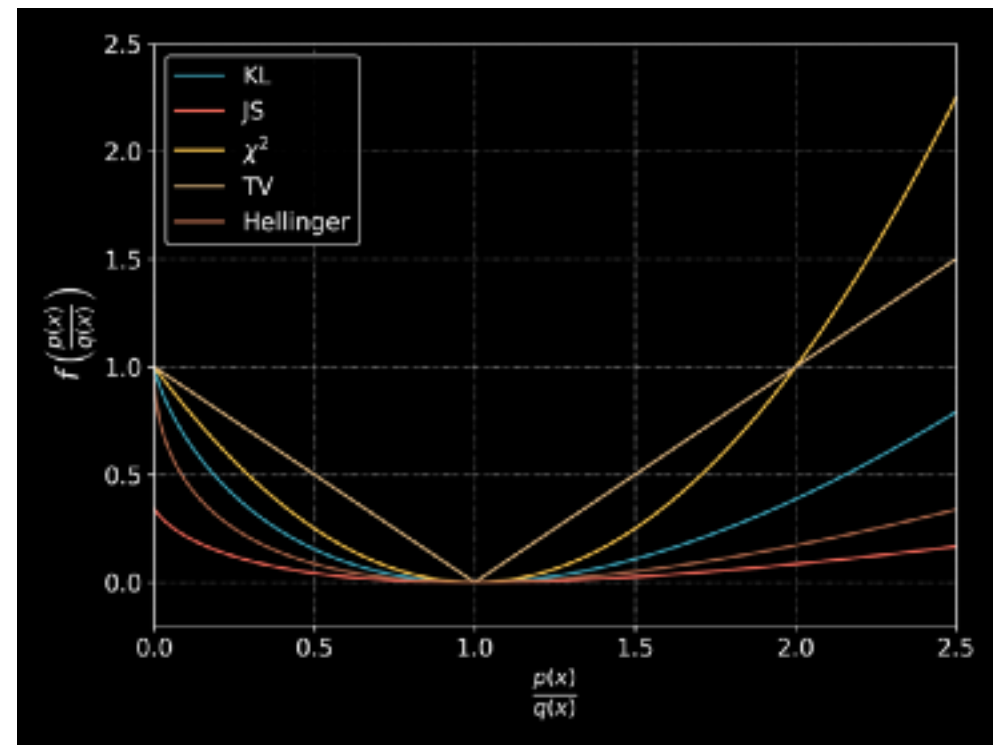
KL Interpolation

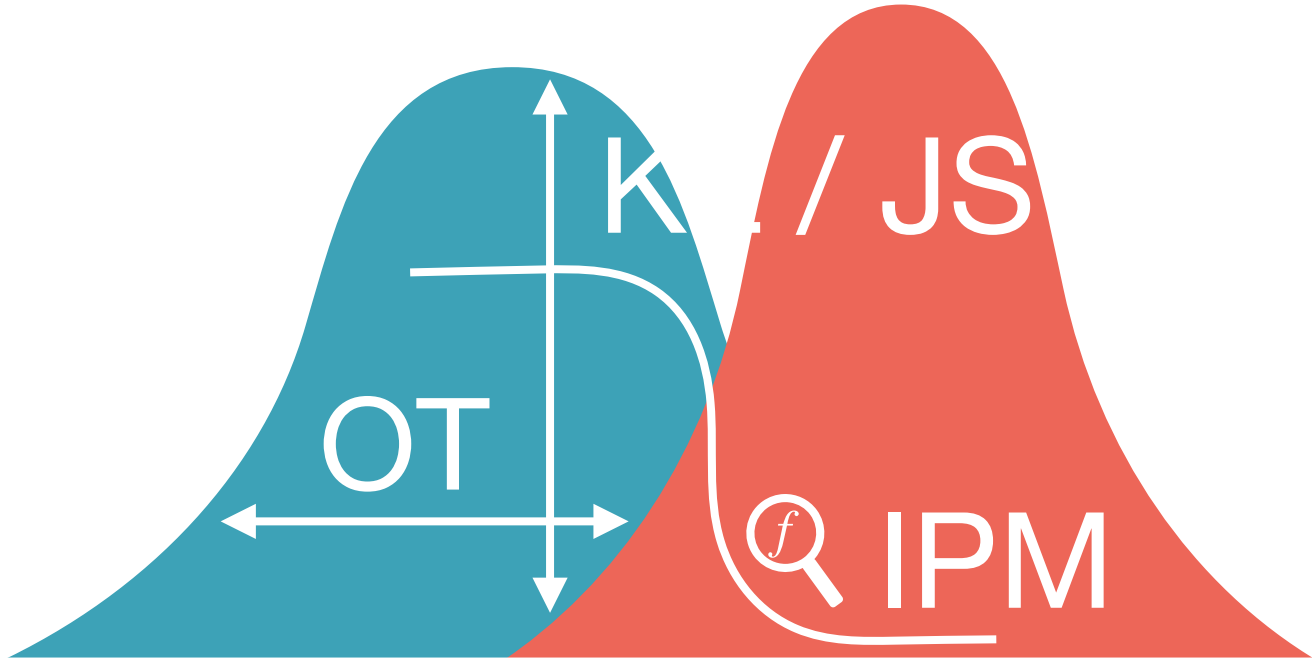


OT Interpolation





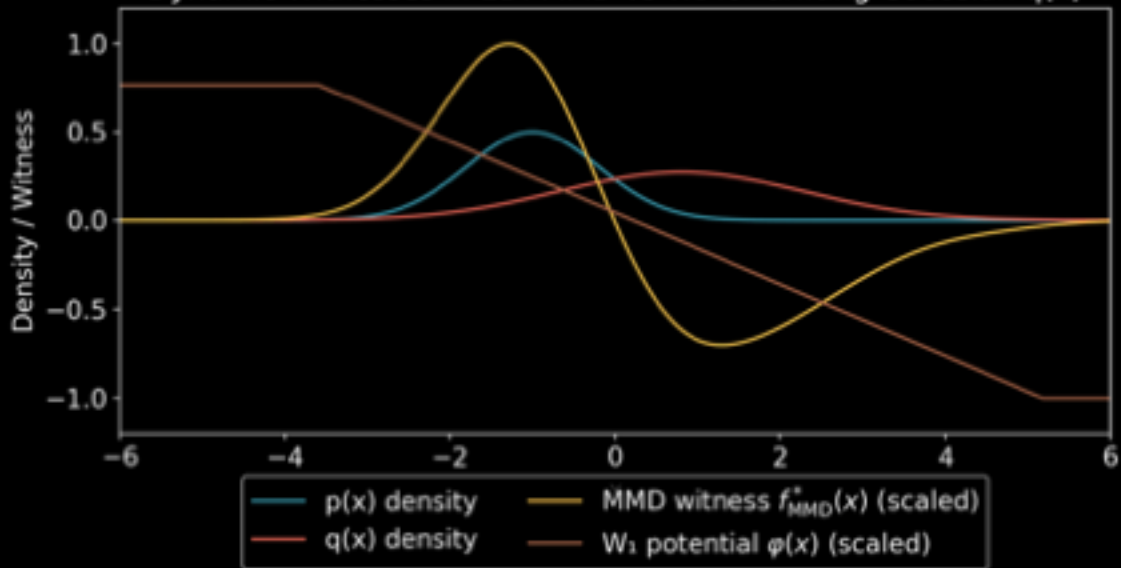




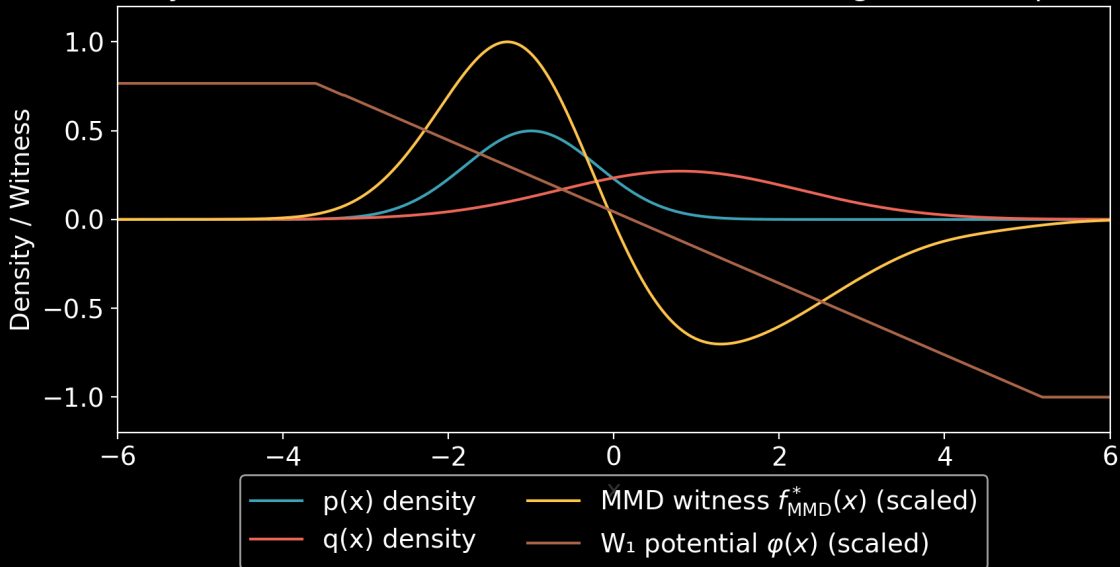


{ geometry, smoothness, and tractability} tradeoff

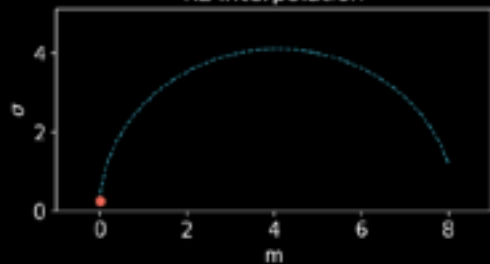
Dynamic Witnesses in 1D: MMD vs W_1 for drifting Gaussian $q(x)$



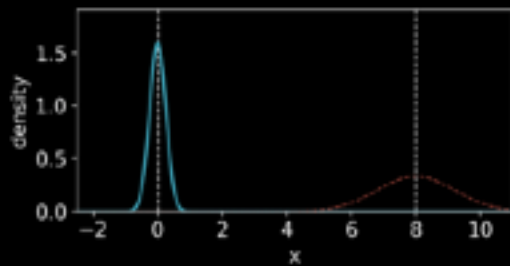
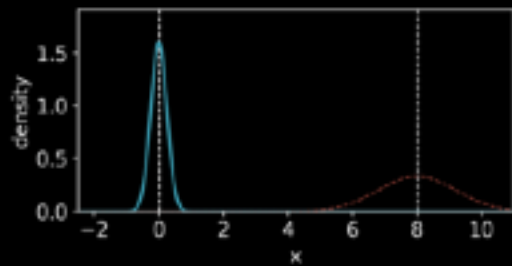
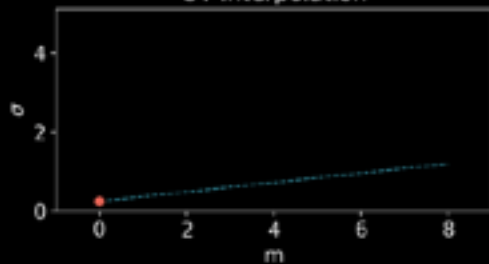
Dynamic Witnesses in 1D: MMD vs W_1 for drifting Gaussian $q(x)$



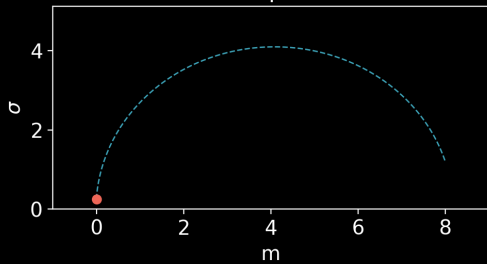
KL Interpolation



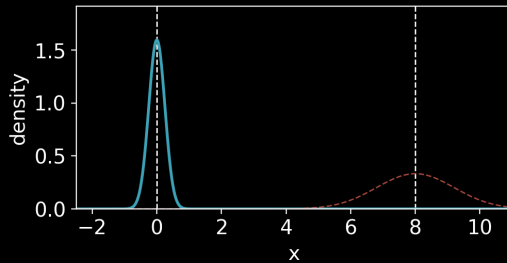
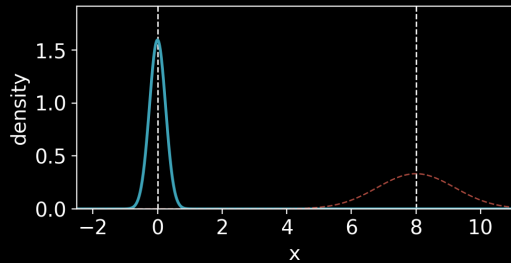
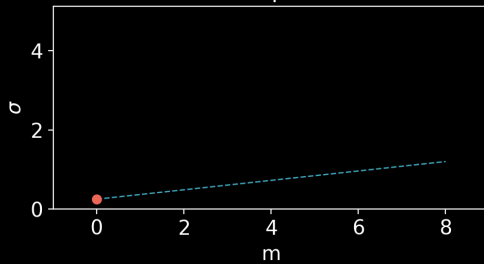
OT Interpolation



KL Interpolation



OT Interpolation

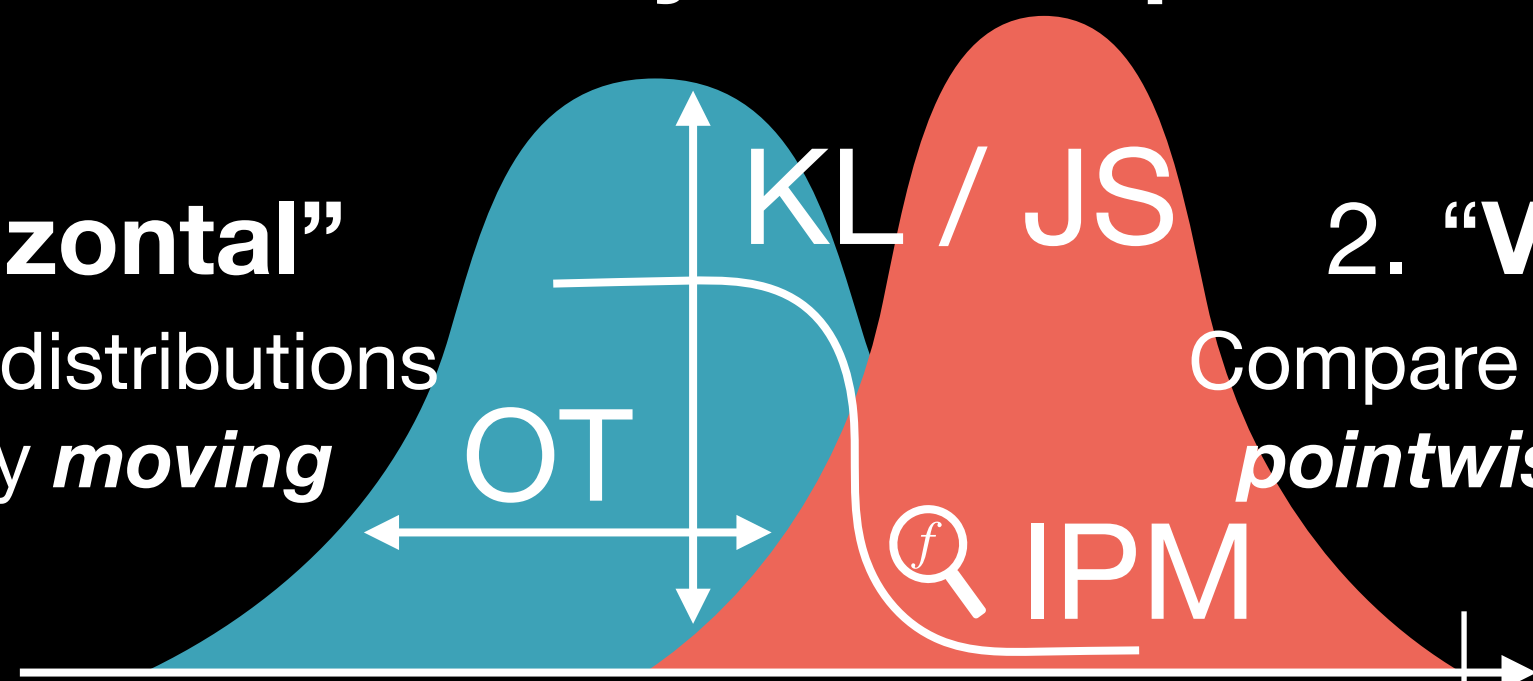


Last Time

Three Ways to Compare

1. “Horizontal”

Compare distributions globally by *moving* mass.



2. “Vertical”

Compare locally via *pointwise* density ratios.

3. “Functional”

Compare via expectation of *witness* functions

3. “Functional” Comparison with IPMs

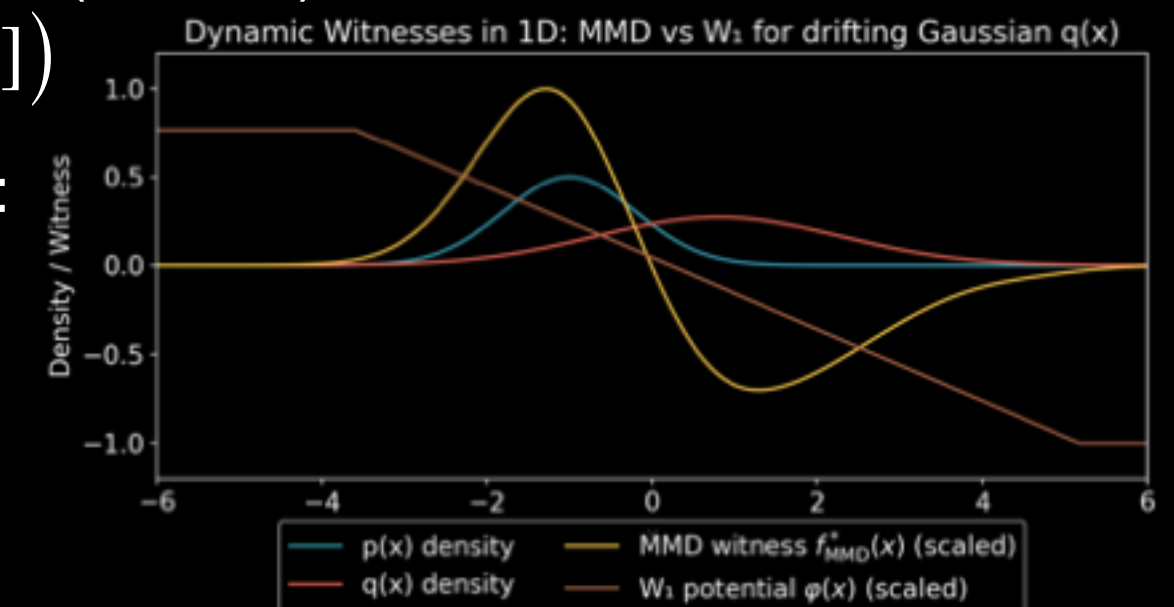
- Compare via expectation gaps of test (witness) functions:

$$d_{\mathcal{F}}(\alpha, \beta) = \sup_{f \in \mathcal{F}} (\mathbb{E}_{\alpha}[f] - \mathbb{E}_{\beta}[f])$$

- Function class $\mathcal{F}$ defines geometry:

- $\|f\|_{\infty} \leq 1 \rightarrow$ Total Variation
- $\text{Lip}(f) \leq 1 \rightarrow$ Wasserstein-1
- $\|f\|_{\mathcal{H}_k} \leq 1 \rightarrow$ MMD

- Witness function highlights where distributions differ.
- Smooth vs rough witnesses $\rightarrow$ different sensitivity / efficiency trade-offs.



2. “Vertical” Comparison with f -divergences

- For convex entropy function f :

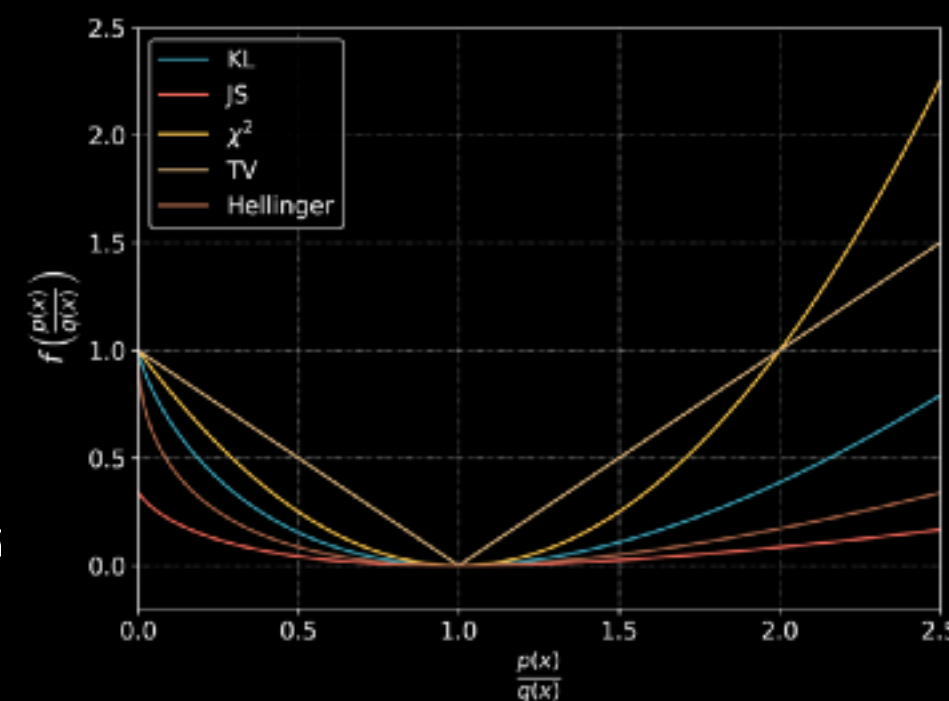
$$D_f(\alpha \| \beta) = \int_X f\left(\frac{d\alpha}{d\beta}(x)\right) d\beta(x)$$

- Choice of f determines form of divergence
- Examples:

$$\text{KLD} : \frac{1}{2} \int \log\left(\frac{d\alpha}{d\beta}\right) d\alpha(x) + \frac{1}{2} \int \log\left(\frac{d\beta}{d\alpha}\right) d\beta(x)$$

$$\text{TV} : \int |d\alpha - d\beta|$$

$$\text{Hellinger} : \int |\sqrt{d\alpha} - \sqrt{d\beta}|^2$$



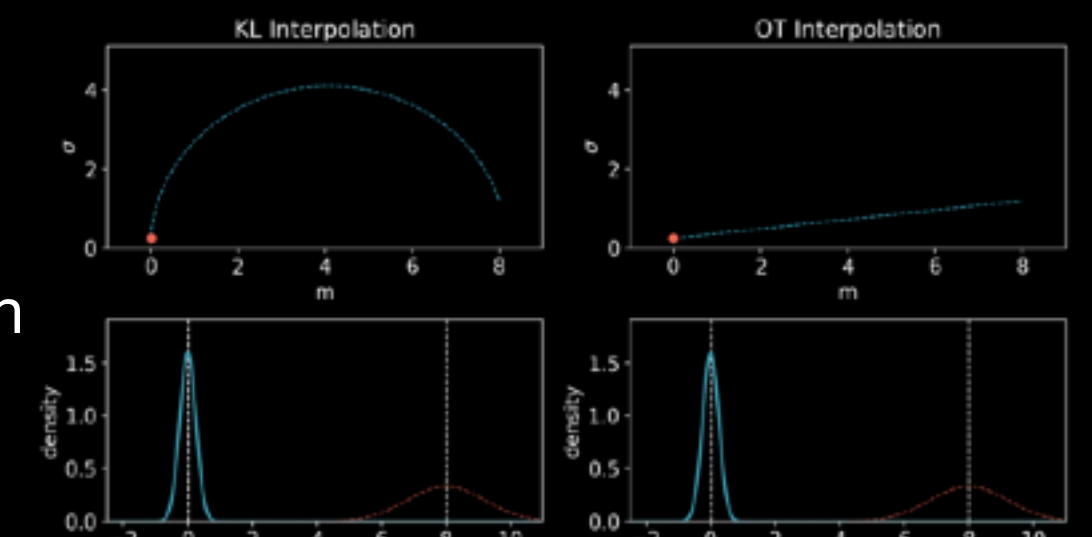
Trade-offs, Comparison & Choice

- Choice of divergence = **{ geometry, smoothness, and tractability } tradeoff:**

- OT: geometry-aware, interpretable, computationally and statistically costly.
- f -divergences: efficient, information-theoretic, need overlapping support.
- IPMs: tunable via $\mathcal{F}$, often dimension-free, but miss fine geometry.

- No one size fits all: choose based on

- overlap vs. disjoint support
- availability of geometric information
- computational constraints



Today